

Requirement Analysis

Human Development Index (HDI) Prediction Using Machine Learning

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Project Title: Requirement Analysis
Project Name: Human Development Index (HDI) Prediction Using Machine Learning
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1. Introduction

Requirement analysis is an important phase in software and data science project development. It helps identify the needs of the users, define the scope of the system, and establish the technical and operational conditions required for successful implementation. In the context of the present project, requirement analysis provides a structured understanding of how the Human Development Index (HDI) prediction system should be designed, developed, and evaluated.

The project focuses on predicting HDI values using machine learning techniques. The system is intended to analyze structured development-related data and generate meaningful predictions that can support academic learning, research, and practical decision-making. A clear requirement analysis is therefore necessary to ensure that the proposed solution is relevant, feasible, and suitable for college-level implementation.

2. Requirement Analysis Overview

Requirement analysis is the process of collecting and documenting the needs that the system must satisfy. It involves understanding the problem domain, identifying user expectations, and defining the features and constraints of the project. For this project, the analysis is based on the need to build a simple, accurate, and interpretable machine learning system for HDI prediction.

The requirement analysis covers both functional and non-functional aspects of the system. It also considers the software, hardware, user, and system requirements needed to implement the project effectively. This ensures that the final solution is practical, manageable, and aligned with the project objectives.

3. Functional Requirements

Functional requirements define what the system must do. The following functional requirements are identified for the HDI prediction project.

ID	Functional Requirement
FR1	The system should accept structured input data related to HDI indicators such as education, income, and life expectancy.

ID	Functional Requirement
FR2	The system should preprocess the dataset by handling missing values and preparing it for model training.
FR3	The system should train a machine learning model to predict HDI values.
FR4	The system should evaluate the model using appropriate performance metrics.
FR5	The system should display prediction results in a clear and understandable format.
FR6	The system should allow users to input data and obtain prediction output.
FR7	The system should support basic visualization of data and results where applicable.
FR8	The system should be simple enough for beginners to understand and use.

4. Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system. These requirements are important for making the project reliable, maintainable, and user-friendly.

Category	Requirement
Usability	The system should be easy to use for students, researchers, and beginners.
Reliability	The system should produce consistent and stable results during normal use.
Performance	The system should respond within an acceptable time for prediction tasks.
Maintainability	The code and project structure should be organized and easy to update.
Scalability	The system should be capable of handling additional data or features in future.
Security	The system should not expose sensitive or unnecessary data.
Portability	The system should be able to run on a standard computer environment with Python.

5. Software Requirements

The software requirements describe the tools and technologies needed to implement the project.

Software Component	Requirement
Programming Language	Python
IDE	Visual Studio Code or Jupyter Notebook
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Web Framework	Flask (optional for deployment)
Dataset Format	CSV file
Operating System	Windows, Linux, or macOS

Software Component	Requirement
Version Control	Git and GitHub

6. Hardware Requirements

The hardware requirements are modest because the project is designed for academic and beginner-level implementation.

Hardware Component	Minimum Requirement
Processor	Dual-core processor or higher
RAM	4 GB or more
Storage	10 GB free disk space
Display	Standard monitor
Internet Connection	Optional, only for downloading packages or datasets

7. User Requirements

The system should meet the requirements of its intended users. These users include students, researchers, developers, and academic evaluators who want to understand and use HDI prediction.

The user requirements are as follows:

- Users should be able to upload or load HDI-related data easily.
- Users should be able to obtain predictions without requiring advanced technical knowledge.
- Users should be able to understand the output in a simple way.
- Users should be able to use the system for academic demonstration and project presentation.
- Users should be able to observe the model's effectiveness through basic evaluation results.

8. System Requirements

The system requirements describe the overall environment and capabilities needed for the successful operation of the project.

- The system should support data loading from a CSV file.
- The system should provide preprocessing features such as handling missing values and feature preparation.
- The system should train a regression-based model suitable for HDI prediction.
- The system should generate predictions and display results clearly.
- The system should run on a standard Python environment without requiring high-end hardware.
- The system should be structured in a way that allows future improvement and extension.

9. Feasibility Analysis

Feasibility analysis helps determine whether the project can be implemented successfully within the available resources and time.

Technical Feasibility

The project is technically feasible because the required tools and libraries are widely available and suitable for machine learning tasks. Python, pandas, and scikit-learn provide the necessary functionality for data processing and model development. The project does not require highly advanced infrastructure.

Economic Feasibility

The project is economically feasible because it can be implemented using open-source tools and standard hardware. There are no significant licensing costs involved, and the required resources are affordable for academic work.

Operational Feasibility

The project is operationally feasible because the system is designed to be simple, understandable, and useful for students and researchers. It can be used for academic purposes, project demonstration, and future portfolio presentation.

10. Assumptions and Constraints

Assumptions

- The dataset used for the project is available in a structured format.
- The project will use a supervised learning approach for HDI prediction.
- The users have basic knowledge of using Python and simple software tools.
- The system will be developed as an academic project with limited resources.

Constraints

- The project may be limited by the size and quality of the available dataset.
- The model accuracy may depend on the quality of the input features.
- The project may require time for data cleaning and model evaluation.
- The deployment feature may be limited depending on available resources.

11. Requirement Summary

In summary, the project requires a machine learning system that can read HDI-related data, preprocess it, train a prediction model, and present the results clearly. The requirements include both functional and non-functional aspects such as usability, performance, maintainability, and scalability. The software and hardware needs are moderate and suitable for an academic project.

The analysis confirms that the project is realistic, practical, and appropriate for a final-year B.Tech implementation. It also provides a clear direction for development, testing, and evaluation.

12. Conclusion

Requirement analysis is a critical step in the successful development of the Human Development Index prediction project. It helps identify the features, constraints, resources, and expectations that guide the development process. By analyzing the functional and non-functional requirements, the project becomes more structured, manageable, and relevant to real-world applications.

The requirement analysis shows that the proposed system is feasible, educational, and suitable for college submission. It provides a strong foundation for implementation and ensures that the final project remains aligned with its academic and practical objectives.